

KOLBA, Russian Federation

WMO: 296750

Lat: 55.1225N

Lon: 93.3837E

Elev: 1565

StdP: 13.88

Time Zone: 7.00 (E07)

Period: 90-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.1	(c) -29.8	(d) -40.2	(e) 0.6	(f) -35.6	(g) -34.0	(h) 0.8	(i) -29.4	(j) 11.9	(k) 17.8	(l) 9.4	(m) 16.0	(n) 0.0	(o) 230	(p) 0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.3	(c) 82.2	(d) 64.4	(e) 78.6	(f) 63.3	(g) 75.3	(h) 61.8	(i) 67.6	(j) 77.5	(k) 65.6	(l) 74.7	(m) 63.8	(n) 72.1	(o) 2.7	(p) 50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	96.0	71.6	62.4	89.2	69.7	60.3	82.8	68.0	33.1	77.7	31.4	75.0	30.0	72.2	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.6	(b) 7.3	(c) 5.5	(e) -40.9	(f) 89.4	(g) 10.0	(h) 2.9	(i) -48.1	(j) 91.5	(k) -53.9	(l) 93.2	(m) -59.6	(n) 94.8	(o) -66.8	(p) 96.9		
(a) 9.6	(b) 7.3	(c) 5.5	(e) -37.8	(f) 71.0	(g) 8.5	(h) 4.5	(i) -43.9	(j) 74.3	(k) -48.8	(l) 76.9	(m) -53.6	(n) 79.5	(o) -59.7	(p) 82.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.9	-0.2	6.5	21.1	33.3	45.6	57.9	61.9	56.7	45.1	33.1	16.8	3.7		
	DBStd	23.73	15.95	13.28	11.62	9.17	8.42	6.54	5.04	6.04	7.01	9.60	13.45	15.16		
	HDD50	7522	1556	1219	897	503	187	13	0	11	179	525	997	1435		
	HDD65	12116	2021	1639	1362	951	602	222	120	266	598	988	1447	1900		
	CDD50	925	0	0	0	1	51	251	370	218	32	2	0	0		
	CDD65	44	0	0	0	0	0	10	25	8	1	0	0	0		
	CDH74	901	0	0	0	0	44	303	375	158	21	0	0	0		
	CDH80	171	0	0	0	0	4	58	68	37	4	0	0	0		
	WSAvg	1.5	1.1	1.5	1.9	2.0	2.1	1.5	1.1	0.9	1.2	1.5	1.4	1.3		
	PrecAvg	22.1	0.7	0.6	0.8	1.6	1.9	2.8	3.1	3.5	2.5	1.9	1.6	1.1		
(15) Precipitation	PrecMax	29.4	1.1	1.3	2.1	3.1	4.2	4.9	6.8	7.0	5.2	3.9	3.8	1.9		
	PrecMin	16.5	0.2	0.2	0.2	0.4	0.7	0.6	0.7	1.7	0.5	0.7	0.7	0.2		
	PrecStd	3.1	0.3	0.3	0.4	0.7	0.8	1.1	1.4	1.3	1.1	0.8	0.7	0.4		
	PrecStd	3.1	0.3	0.3	0.4	0.7	0.8	1.1	1.4	1.3	1.1	0.8	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	38.0	54.3	67.4	82.1	85.9	86.4	84.9	76.5	63.0	47.4	33.8		
		MCWB	30.4	31.1	40.6	48.1	59.0	63.1	68.0	68.6	59.5	48.6	39.6	29.2		
	2%	DB	29.6	32.4	46.7	59.2	75.1	81.7	81.8	78.1	67.3	55.7	41.6	29.6		
		MCWB	26.3	27.7	36.5	44.4	55.6	63.5	65.4	64.7	55.4	46.1	36.0	26.4		
	5%	DB	25.7	29.5	41.7	53.3	70.0	77.8	78.3	73.8	61.3	50.7	37.1	27.4		
		MCWB	22.9	25.6	33.9	41.9	53.4	61.9	64.7	63.0	52.6	44.3	33.3	24.7		
	10%	DB	21.6	26.0	37.3	47.9	64.8	73.7	75.2	69.3	56.8	46.4	33.9	24.7		
		MCWB	19.3	22.7	31.5	39.0	50.8	60.4	63.1	61.0	50.3	41.3	31.1	22.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.8	31.9	40.7	49.8	61.1	68.3	70.5	70.3	61.9	50.9	41.1	29.4		
		MCDB	33.1	36.1	53.1	61.1	74.7	77.8	80.1	79.2	72.3	59.2	46.3	32.4		
	2%	WB	26.7	28.4	37.1	46.3	57.6	65.6	68.4	66.4	56.6	47.6	36.7	26.8		
		MCDB	28.7	32.0	45.2	57.9	72.3	76.8	78.4	75.6	64.5	53.6	40.9	29.3		
	5%	WB	22.5	25.5	34.5	42.7	55.0	63.8	66.5	64.0	53.9	44.8	33.7	24.9		
		MCDB	25.5	29.4	40.8	52.6	67.0	74.3	75.2	72.1	60.0	49.9	36.5	27.3		
	10%	WB	19.4	22.8	31.9	39.7	51.9	61.5	64.6	61.6	51.3	42.1	31.2	22.6		
		MCDB	21.5	26.0	36.8	47.2	62.7	71.5	72.2	68.4	55.8	46.1	33.9	24.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	21.7	24.3	21.7	23.5	23.7	21.3	19.7	18.9	16.6	15.2	15.9		
		MCDBR	15.7	21.0	30.1	31.8	36.2	33.8	28.9	29.1	29.8	25.5	16.3	15.4		
		MCWBR	13.5	16.9	20.8	19.0	19.9	17.7	14.9	16.7	17.4	15.9	11.3	13.4		
	5% WB	MCDBR	15.9	19.4	27.5	28.8	29.6	27.3	25.1	26.2	24.9	23.4	14.6	16.1		
		MCWBR	14.0	15.9	19.1	18.2	17.9	15.9	14.8	16.3	15.9	15.3	11.1	14.5		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.310	0.355	0.394	0.409	0.399	0.414	0.400	0.362	0.343	0.311	0.279		
	taud		2.046	2.014	1.964	2.099	2.193	2.327	2.307	2.314	2.367	2.263	2.070	1.981		
	Ebn at Noon		197	233	250	260	264	269	261	258	252	224	178	159		
	Edh at Noon		21	32	43	44	42	37	38	35	29	25	21	18		
(44) All-Sky Solar Radiation	RadAvg		285	604	1026	1349	1571	1874	1655	1360	906	493	252	178		
	RadStd		31	40	94	84	182	169	122	125	116	45	33	21		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page